

Bound-Covering Subcases of the Asplund Product Problem for $C_k(X)$

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Status

This is a claimed partial result, subject to human review. It proves structured-factor subcases of Problem 3.4 from Fabian–Kakol–Leiderman, *Asplund spaces $C_k(X)$ beyond Banach spaces*, arXiv:2510.01873. It is not a full solution of the source problem.

Source Question

The source paper proves that, for a Tychonoff space X , the lcs $C_k(X)$ is Asplund if and only if every compact subset of X is scattered. Immediately after Remark 3.3, which covers the cases of an Asplund Banach factor and a second $C_k(Y)$ factor, the paper asks:

Problem 3.4. Assume that $C_k(X)$ is an Asplund space. Is the product $C_k(X) \times E$ Asplund for every Asplund lcs E ?

Remark 3.3. (a) Statement (1) in Theorem 3.2 can be strengthened to: ‘For every Asplund Banach space E the product $C_k(X) \times E$ is also Asplund.’

Indeed, in the proof of ‘(2) $\implies$ (1)’, it is enough to replace, in suitable places, ‘ $C_k(X)$ ’ by ‘ $C_k(X) \times E$ ’. More specifically, the mapping $C_k(X) \times E \ni (f, g) \longmapsto (f|_L, g) \in C(L) \times E$ is still ‘bound covering’ (as it was the mapping T). And the latter product (of two Banach spaces) is Asplund by [29]. Another way how to check this fact is by using the paper [8];

(b) From another hand, if both spaces X and Y have all compact subsets scattered, respectively, then $C_k(X) \times C_k(Y)$ is Asplund. Indeed, it is well known that $C_k(X) \times C_k(Y)$ is isomorphic to $C_k(X \oplus Y)$, and then in the direct sum $X \oplus Y$ all compact subsets are scattered and Theorem 3.2 applies.

Problem 3.4. Assume that $C_k(X)$ is an Asplund space. Is the product $C_k(X) \times E$ Asplund for every Asplund lcs E ?

Analyzing the proof of ‘(2) $\implies$ (1)’ in Theorem 3.2, we can deduce the following fact related with Theorem 3.10: see also [8, Theorem 3.2]

Why This Is Only Partial

Taking X to be a singleton identifies $C_k(X)$ with $\mathbb{R}$. Thus the source problem contains the basic open question whether $E \times \mathbb{R}$ is Asplund for every Asplund lcs E . Kakol–Leiderman, arXiv:2412.10221, Question 3.1, lists that one-dimensional product problem as open. The theorem below therefore restricts E to locally convex spaces whose topology is controlled by open bound-covering quotient maps onto Asplund Banach spaces.

Definitions Used

A locally convex space Z is called Asplund if every continuous convex real function on every nonempty open convex subset of Z is Frechet differentiable at the points of a dense G_δ subset of its domain. Differentiability is understood in the usual locally convex sense: the error is uniform on bounded sets.

For compact $L \subset X$, write

$$V_L^\varepsilon = \{f \in C_k(X) : |f(t)| < \varepsilon \text{ for all } t \in L\}.$$

Following Eyland–Sharp, as recalled in Kakol–Leiderman arXiv:2412.11510, a lcs E is called bound covering if it has a directed defining family $\mathcal{P}$ of continuous seminorms such that, for each $p \in \mathcal{P}$, the quotient map $q_p : E \rightarrow E_p = (E/\ker p, \bar{p})$ is open and every bounded set in E_p is contained in the image under q_p of a bounded set in E . In this packet we only use openness and the fact that the spaces E_p are Banach Asplund spaces.

Proof Intuition

The proof combines compact restriction in the $C_k(X)$ coordinate with Banach quotient reduction in the E coordinate. Local boundedness of a continuous convex function on $C_k(X) \times E$ is witnessed by a neighborhood determined by one compact set $L \subset X$ and one quotient seminorm p on E . Convexity then forces the function to be constant on the kernel of the map

$$(f, e) \mapsto (f|_L, q_p(e)).$$

Thus the function factors through the Banach space $C(L) \times E_p$. Since $C_k(X)$ is Asplund, L is scattered and $C(L)$ is Asplund; by hypothesis E_p is Asplund. Frechet differentiability in this Banach quotient pulls back uniformly on bounded sets.

Lemma 1 (bounded convex functions on affine fibers). *Let φ be convex on a convex set D in a real vector space. Suppose $z \in D$, d is a vector, and $z + \lambda d \in D$ for all $\lambda \in \mathbb{R}$. If φ is bounded above on the affine line $z + \mathbb{R}d$, then $\varphi(z + d) = \varphi(z)$.*

Proof. For $\lambda > 1$, convexity gives

$$\varphi(z + d) \leq (1 - \lambda^{-1})\varphi(z) + \lambda^{-1}\varphi(z + \lambda d).$$

Letting $\lambda \rightarrow \infty$ and using boundedness above on the line yields $\varphi(z + d) \leq \varphi(z)$. Applying the same argument to $z + d$ and $-d$ gives the reverse inequality. $\square$

Theorem 1. *Let X be Tychonoff and assume that $C_k(X)$ is an Asplund locally convex space. Let E be a bound-covering lcs with a directed defining family $\mathcal{P}$ such that each Banach quotient $E_p = (E/\ker p, \bar{p})$, $p \in \mathcal{P}$, is Asplund. Then*

$$C_k(X) \times E$$

is an Asplund locally convex space.

Proof. Let $Z = C_k(X) \times E$, let $D \subset Z$ be nonempty, open, and convex, and let $\varphi : D \rightarrow \mathbb{R}$ be continuous and convex. We prove that φ is Frechet differentiable on a dense G_δ subset of D .

Choose $u = (f_0, e_0) \in D$. By continuity there is a basic convex neighborhood of u on which φ is bounded above and which is contained in D . Thus there are a compact set $L \subset X$, a seminorm $p \in \mathcal{P}$, and $\varepsilon > 0$ such that

$$u + (V_L^\varepsilon \times q_p^{-1}(\varepsilon B_{E_p}))$$

is contained in D and φ is bounded above on it.

Define

$$Q : Z \longrightarrow C(L) \times E_p, \quad Q(f, e) = (f|_L, q_p(e)).$$

The map Q is continuous, open, and surjective. For the restriction coordinate this is the Tietze–Urysohn extension lemma used in arXiv:2510.01873: a small function on L extends to a function on X with small global range, hence small on every compact set appearing in a basic $C_k(X)$ -neighborhood. For the E coordinate this is the assumed openness of q_p . Hence $Q[D]$ is an open convex subset of the Banach space

$$B := C(L) \times E_p.$$

Let $N = \ker Q$. The chosen neighborhood of u contains the full affine space $u + N$, and φ is bounded above on $u + N$. We claim that D is saturated by N and that φ is constant on each N -fiber inside D . Fix $z \in D$ and $d \in N$. Choose a neighborhood $O_z \subset D$ of z on which φ is bounded above. For each real t , choose $\lambda > 1$ so large that

$$w_\lambda = \frac{z - \lambda^{-1}u}{1 - \lambda^{-1}} \in O_z.$$

Then

$$z + td = \lambda^{-1}(u + \lambda td) + (1 - \lambda^{-1})w_\lambda \in D,$$

because $u + \lambda td \in u + N \subset D$ and $w_\lambda \in D$. The same convex-combination estimate, using the upper bounds on $u + N$ and O_z , shows that φ is bounded above on the whole line $z + \mathbb{R}d$. By Lemma 1, $\varphi(z + d) = \varphi(z)$. Thus D is N -saturated and φ is fiber-constant.

Thus there is a well-defined convex function $\psi : Q[D] \rightarrow \mathbb{R}$ such that $\varphi = \psi \circ Q$ on D . It is continuous: around each point of $Q[D]$, the openness of Q , the continuity of φ , and the fiber-constancy just proved give a Banach-space neighborhood on which ψ is bounded above. A convex function locally bounded above on a Banach-space open set is continuous.

Since $C_k(X)$ is Asplund, Theorem 3.2 of arXiv:2510.01873 implies that every compact subset of X is scattered. Hence L is scattered, and the Namioka–Phelps theorem gives that $C(L)$ is an Asplund Banach space. The space E_p is Asplund by hypothesis, so B is an Asplund Banach space. Therefore ψ is Frechet differentiable on a dense G_δ subset Ω of $Q[D]$.

Because Q is continuous and open, $Q^{-1}(\Omega) \cap D$ is a dense G_δ subset of D . Let $z \in Q^{-1}(\Omega) \cap D$ and let $\xi \in B^*$ be the Frechet derivative of ψ at Qz . We show that $\xi \circ Q$ is the Frechet derivative

of φ at z . If $M \subset Z$ is bounded, then $Q[M]$ is bounded in B : bounded sets in $C_k(X)$ restrict to bounded sets in $C(L)$, and continuous linear maps send bounded sets to bounded sets. Hence

$$\sup_{h \in M} |\varphi(z + th) - \varphi(z) - t\xi(Qh)| = \sup_{h \in M} |\psi(Qz + tQh) - \psi(Qz) - t\xi(Qh)| = o(t)$$

as $t \rightarrow 0$, with the usual restriction to sufficiently small t so that $z + tM \subset D$. This is Frechet differentiability of φ at z .

Therefore every continuous convex φ on every nonempty open convex $D \subset Z$ has a dense G_δ set of Frechet differentiability points. Thus $Z = C_k(X) \times E$ is Asplund. $\square$

Concrete Consequences

The theorem applies to the following classes of Asplund factors E .

- If $E = \prod_{\alpha \in A} E_\alpha$ is a locally convex product of Asplund Banach spaces, take $\mathcal{P}$ to be the finite-coordinate maximum seminorms. The quotient levels are finite products of the E_α 's.
- The same argument applies to the zero-basepoint Σ -product of a family of Asplund Banach spaces, using finite-support lifts for the quotient maps.
- If E is equipped with a weak topology, the one-functional quotient levels are one-dimensional. Thus $C_k(X) \times E_w$ is Asplund whenever $C_k(X)$ is.
- If E is an Asplund quojection, then arXiv:2412.11510, Theorem 5.2, shows that its Banach levels are Asplund and its quotient maps are bound-covering. Hence $C_k(X) \times E$ is Asplund. This includes reflexive quojections.

Verification and Novelty Notes

The proof uses the factorization proof of Theorem 3.2 and Remark 3.3 in arXiv:2510.01873, the finite-coordinate product mechanism and Σ -product analogue from arXiv:2412.10221, and the bound-covering quotient framework recalled in arXiv:2412.11510. The product paper also records, in Question 3.1, that the arbitrary one-dimensional Asplund product problem remains open; this is why the present theorem is classified as a partial result.

The local run indexes were searched for arXiv:2510.01873, the source title, and phrases involving $C_k(X)$, Asplund products, Banach products, and Σ -products. No exact duplicate packet was found. No fresh external web search was performed for this partial theorem.

Human Review Recommendation

Review should focus on the kernel-saturation step and on the use of openness of the quotient map q_p to make $Q[D]$ a Banach-space open domain. If those are accepted, the remainder is a standard pullback of Frechet differentiability through a continuous bounded-set-preserving linear map.

References

- [1] M. Fabian, J. Kakol, and A. Leiderman, *Asplund spaces $C_k(X)$ beyond Banach spaces*, arXiv:2510.01873.

- [2] J. Kakol and A. Leiderman, *On the product of Weak Asplund locally convex spaces*, arXiv:2412.10221.
- [3] J. Kakol and A. Leiderman, *Asplund locally convex spaces and the finest locally convex topology*, arXiv:2412.11510.
- [4] I. Namioka and R. R. Phelps, *Banach spaces which are Asplund spaces*, Duke Math. J. 42 (1975), 735–750.